

Quantum Game Jam: Problem Sheet

Select exactly one track. Wrap the game entirely around the chosen quantum principle.

Problem Tracks

Track 1: Basis Switching & Measurement Scrambling

EASY

- **Physical Principle:** In quantum mechanics, observing a system requires picking a measurement basis. A qubit in state $|0\rangle$ or $|1\rangle$ measured in the computational (Z) basis yields deterministic outcomes.
- **The Core Mechanism:** Measuring that exact same state in the complementary diagonal (X) basis ($|+\rangle$, $|-\rangle$) forces an irreversible, uniform 50/50 collapse. The original state information is permanently erased simply by inspecting it along the incompatible axis.
- **Key Formalism:** Conjugate measurement bases, projection operators, state collapse, and basis incompatibility.

Track 2: The Deutsch–Jozsa Parity Oracle

HARD

- **Physical Principle:** A black-box function $f(x)$ mapping binary strings to $\{0, 1\}$ is guaranteed to be either *constant* (all outputs identical) or *balanced* (outputs split evenly 50/50). Classically, determining this property requires up to $2^{n-1} + 1$ individual queries in the worst case.
- **The Core Mechanism:** Using a superposition of all possible inputs and phase kickback ($U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$), a quantum algorithm exploits multi-path interference. Destructive interference cancels all paths if the function is balanced, allowing the global property to be evaluated in exactly one single computational query.
- **Key Formalism:** Phase kickback, Walsh-Hadamard transformation, and quantum computational speedup via global parity interference.

Track 3: Unitary Quantum Gates & Reversible Flow

EASY

- **Physical Principle:** Classical logic operations (AND , OR) are non-invertible and erase input history. In contrast, all unmeasured quantum logic operations are unitary transformations ($U^\dagger U = I$), meaning evolution is strictly reversible and preserves state length on the Bloch sphere.
- **The Core Mechanism:** Elementary single-qubit transformations—the Bit Flip (X), the Phase Flip (Z), and the Hadamard (H) gate—rotate quantum state vectors deterministically. Any gate sequence can be precisely unwound or traversed in reverse.
- **Key Formalism:** Bloch sphere coordinates, single-qubit unitary operators, reversibility, and state vector trajectories.

Track 4: Grover's Amplitude Amplification

HARD

- **Physical Principle:** Finding a marked target among N unsorted items classically requires $O(N)$ linear evaluations. Quantum systems can identify the target in $O(\sqrt{N})$ operations without sequential testing.
- **The Core Mechanism:** A two-step iterative cycle: (1) An oracle introduces a phase inversion (-1 sign flip) specifically to the marked target state; (2) A diffusion operator reflects all amplitudes across their mathematical mean ($2|s\rangle\langle s| - I$). This constructive-destructive cycle amplifies the target state's amplitude while suppressing all non-target states. Repeated application beyond the optimal number of iterations causes over-rotation and degrades accuracy.
- **Key Formalism:** Oracle phase mark, geometric inversion about the mean, state vector amplitude reflection, and rotation angle cycles.

Track 5: Superdense Coding

EASY

- **Physical Principle:** Holevo's theorem establishes that an isolated single qubit can convey at most one bit of classical information upon measurement. However, pre-shared entanglement breaks this transmission constraint.
- **The Core Mechanism:** Two parties share an entangled Bell pair: $|\Phi^+\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$. By performing one of four local unitary operations (I, X, Z, XZ) on solely their own local qubit and sending it to the receiver, the sender transmits two full classical bits of information ($00, 01, 10, 11$) through the transfer of a single physical qubit.
- **Key Formalism:** The 4 Bell states, local Pauli transformations on entangled sub-systems, and Bell-basis measurement decoding.

Track 6: Quantum Error Syndromes & Parity Probes

HARD

- **Physical Principle:** Environmental noise causes bit flips (X) and phase flips (Z) on delicate quantum states. The No-Cloning Theorem strictly prevents backing up states via classical triple copying ($|\psi\rangle \rightarrow |\psi\psi\psi\rangle$).
- **The Core Mechanism:** Data is encoded non-locally across an entangled multi-qubit register. Auxiliary "ancilla" qubits interact with the data register via controlled entangling gates ($CNOT$) to perform indirect parity checks (syndrome measurements). Measuring only the ancilla qubits reveals the location and nature of an error without ever measuring, projecting, or destroying the arbitrary superposition of the protected data state.
- **Key Formalism:** 3-qubit bit/phase flip codes, ancilla-based syndrome extraction, non-destructive parity measurement, and stabilizer corrections.

Submission Deliverables Checklist:

Every team must submit: (1) Chosen track identification; (2) Playable game link or executable; (3) Public repository link; (4) A short video of the team playing the game and showing the core mechanic; (5) Short write-up in the README describing what the team learned about the quantum topic during the first 3 hours and how that concept acts as the central mechanism of the game.